Invariant Structure Representation for Remote Sensing Object Detection Based on Graph Modeling

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***Abstract*— Due to the characteristics of vertical orthophoto imaging, the apparent structural features of the object in the remote sensing (RS) image are relatively stable, such as the cross-shaped structure of the aircraft and the rectangular structure of the vehicle. Compared with the traditional visual features, using these features is conducive to improving the accuracy of object detection. However, there are few studies on such characteristics. In this article, we systematically study the invariant structural features of remote sensing objects and propose a graph focusing aggregation network (GFA-Net) to represent the structural features of remote sensing objects. Among them, in view of the problem that traditional convolutional neural networks (CNNs) are sensitive to the changes in rotation, scale, and other factors, which makes it difficult to extract structural features, we propose the graph focusing process (GFP) based on the idea of graph convolution. Analysis and experiments show that graph structure has significant advantages over Euclidean feature space under CNN in expressing such structural features. In order to realize the end-to-end efficient training of the above model, we design a graph aggregation network (GAN) to update the weight of nodes. We verify the effectiveness of our method on the proposed multitask datasets aircraft component segmentation dataset (ACSD) and the large-scale Fine-grAined object recognItion in high-Resolution RS imagery (FAIR1M). Experiments conducted on the object detection datasets of large-scale Dataset for Object deTection in Aerial images (DOTA) and HRSC2016 prove that the proposed method is superior to the current state-of-the-art (SOTA) method.**

***Index Terms*— Convolutional neural network (CNN), graph modeling, invariant structural feature, object detection, remote sensing (RS).**

I. Introduction

OBJECT detection in remote sensing (RS) scenery is a fundamental yet challenging task [1]–[3]. Its purpose is to locate and recognize objects of predefined categories in any given image. In recent years, with the rapid development of convolutional neural network (CNN), it has brought considerable performance improvements to object detection task in the field of remote sensing [4]–[12]. Different from the natural scene, the object in the remote sensing image shows a relatively stable shape from the bird's eye view due to its special rigid body structure [13]–[15]. Some remote sensing objects (aircraft, vehicles, ships, and so on) are made up of a set of components that have the same shape and fixed relative positions. These apparent structural features can be determined as a strong basis for distinguishing the remote sensing objects (see Fig. 1). Different from the existing rotation-invariant features [16], [17], the invariant structural features contain more complex transformation methods (rotation, translation, scale, and so on), and the applicable objects are relatively more regular (rigid body). The effective extraction and utilization of the structural features of remote sensing objects are of great help in downstream tasks, such as precise localization and fine-grained classification [18], [19].

In feature space extracted by CNN, the features are arranged in the form of a 2-D grid, which determines its relatively good translation invariance but poor rotation invariance and scale invariance [17]. Structural features are required to be preserved as much as possible during translation, rotation, and scaling, which is contrary to the extraction process of CNN. In addition, there is a certain degree of feature misalignment

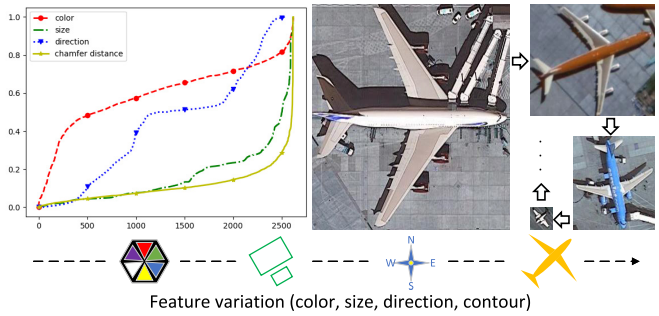


Fig. 1. We filter the aircraft in the DOTA dataset and quantify their visual features (including color, size, direction, and contour). A conclusion can be drawn that, as the number of samples increases, the changes in various visual features gradually stabilize. Changes among contours are relatively minimal and smooth, and the extreme values are relatively limited. The above conclusions are equally applicable to other categories. It can be seen that the contour feature is more representative of remote sensing objects than other features, which we call the invariant structural feature.

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problem between the horizontal RoI and the object in arbitrary orientation [16], and between the RoI feature and the edge details of the rigid object, which is rooted in the modeling space and object representation. The abovementioned problems both may introduce noise features in the process of extracting and decoding structural features of remote sensing objects, thereby affecting the accuracy of their localization and classification. Many efforts [16], [17], [20], [21] have been devoted to alleviating the abovementioned contradictions, but the effect is relatively limited.

Compared with traditional convolutional networks that deal with Euclidean spatial features, graph networks have more natural advantages in modeling structural features. First, the graph structure has inherent rotation invariance, which simply solves the problem of the corresponding rotation feature misalignment problem. Second, the graph structure relies on the edges to establish the connection state between any nodes, rather than the fixed pixel connection mode under CNN, which is suitable for remote sensing objects (aircraft, vehicles, ships, and so on) with general rigid structures. Finally, the graph structure can realize the similar expression of object characteristics at different scales by controlling the number of nodes so that it has scale invariance. It is worth noting that, in remote sensing scenes, the internal relations between the contours of objects and their components are more regular. The model can further improve the efficiency of discrimination through the information interaction between structural features.

However, the introduction of graph network modeling such features for remote sensing object detection tasks still has problems. One is the mismatch of feature structure. The output feature of the backbone network stacked by the convolution operation is a set of gridded data structures in the Euclidean space, which cannot be directly input into the graph network for processing. In response to this problem, we adopt the idea of extracting key points combined with the characteristics of remote sensing objects and propose the graph focusing process (GFP) to complete the above conversion. The second is the problem of graph network layer design. How to deal with the feature interaction between nodes in the graph structure and obtain valuable features is the purpose of the graph network. Inspired by the idea of combining components to assemble a remote sensing object, we design the graph aggregation network (GAN). It can effectively measure the importance of each node and integrate the characteristic information between each node. We integrate the above two innovations and instantiate a multitask detection framework, namely, the graph focusing aggregation network (GFA-Net). It can simultaneously process multiple remote sensing vision subtasks (rotating object detection, component segmentation, and fine-grained classification) and output the corresponding results.

To further explore the potential of the proposed model, we make a dataset for the task of aircraft component segmentation and conduct comparative experiments. The aircraft component segmentation dataset (ACSD) adopts two kinds of labels for the aircraft, which are based on its type and component type. As we all know, aircraft and other remote sensing objects have the characteristics of large interclass differences and small intraclass differences. In recent years, the effect of fine-grained classification of airplanes is often unsatisfactory. The characteristics, position, and even quantity of aircraft components may provide a powerful basis for their recognition, which also inspired us to make this dataset. The model that we propose can make full use of these aircraft components' features, modeling the feature relationship between each component through graph structure, and, finally, output more accurate predictions reasonably.

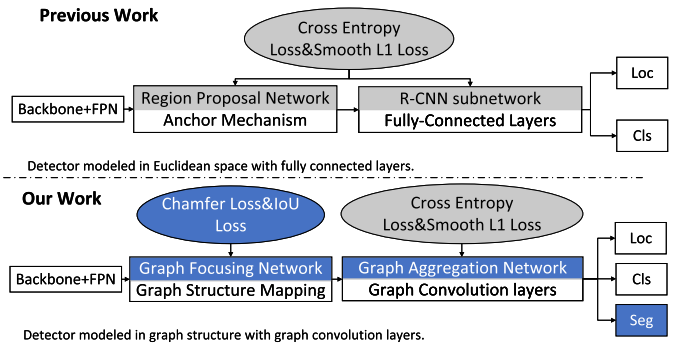


Fig. 2. Differences between the previous work and our work. In the previous work, the detectors are usually modeled in the Euclidean space, applying the region proposal network (RPN) and fully connected layers. In our work, the detector is modeled in a graph structure, applying the graph focusing network (GFN) and the proposed GAN.

Our innovations and main contributions can be summarized as follows:

1) We systematically study the invariant structural characteristics of remote sensing objects and propose a structural feature used to represent remote sensing objects. After a comparative analysis, the graph structure has a significant advantage over the Euclidean feature space under CNN in expressing this structural feature.
2) Aiming at the abovementioned structural feature extraction problem, a graph network model named GFA-Net is proposed. Specifically, we apply a graph structure named GFP to model remote sensing objects and design a graph network called GAN to aggregate features reasonably.
3) Based on the above model, a framework for object fine-grained recognition and detection is proposed. The performance on the rotating object detection task is excellent, reaching the level of state-of-the-art (SOTA) on public remote sensing datasets, such as Fine-grAined object recognItion in high-Resolution RS imagery (FAIR1M), large-scale Dataset for Object deTection in Aerial images (DOTA), and HRSC2016.

The rest of this article is organized as follows: In Section II, the related works are described. We then introduce our method in Section III. The results and analysis of the experiments are presented in Section IV. Finally, we summarize our work in Section V.

II. Related Works

A. Invariant Structural Features Extraction for Remote Sensing Object

The current mainstream remote sensing object detection models can be divided into the following two types according

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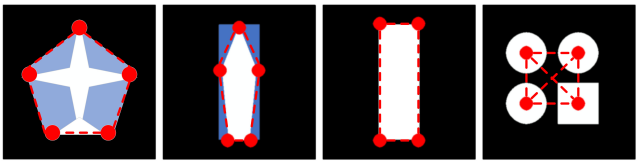


Fig. 3. Apparent structure of the remote sensing object, which is the simplified description of the invariant structural feature.

to their invariant structural features extraction methods for remote sensing objects.

1) Structural Modular Design: Some methods are dedicated to designing targeted structural modules to help the model better extract and characterize the structural features of remote sensing objects. They usually need to regress five parameters (cx, cy, h, w, θ), in which the angle information is used to offset the obtained axis-aligned bounding box. The offset angle is customized clockwise or counterclockwise based on the x-axis direction in polar coordinates. Early work [16], [17], [22]–[24] attributes the distortion of the structural feature to the extraction of the Region-of-Interest (RoI) part and designs a feature alignment RoI extraction module adapted to the rotating object to replace the original structure. The fully convolutional model proposed later [20], [21], [25] also designs a kind of feature alignment module to be connected in series in the head structure, achieving a higher level of performance and resource occupancy. However, on the one hand, the structural module methods rely much on the manual design conception, and on the other hand, it introduces nonnegligible additional computing resource occupancy.

2) Object Representation Modeling: Another stream of methods believes that the introduction of angle information has caused a contradiction with the Euclidean spatial feature representation constructed by CNN, so the quadrilateral rotating object representation method is applied. They adopt corner offsets, key-point heat maps, and so on to establish a mapping of the corners of a rotating object, thereby representing the rotating bounding box. Some works [26], [27] establish the mapping by regressing the offset representation of the object's corner points concerning its midpoint coordinates. The commonly applied smooth L1 loss requires a fixed parameter learning order, while the symmetrical rigid structure of the remote sensing objects blurs its corner order definition, hindering the model learning [28], [29]. [29] designs an automatic organized point set mechanism, avoiding the definition of corner points order by regressing the distribution of a set of redundant points, deriving precise localizing results. Other works [30]–[32] apply the classification method to establish a feature map distinguished by foreground and background, and extract the features of an object into the RoI features at the pixel level. The abovementioned object representation modeling methods based on the Euclidean space tend to have better localization accuracy but perform poorly on the object classification subtask.

B. Graph Structure Modeling for Natural Scene

The graph structure feature modeling is first applied to extract features of the data structures under non-Euclidean space. In recent years, with the rapid development of graph neural networks, converting images into graph structures and then feeding them into graph neural networks are gradually matured. In this section, we first introduce the achievements of the graph neural networks in the natural scene visual domain and then analyze its advantages and challenges under remote sensing scenes.

1) Achievements in Computer Vision: In recent years, the graph network has performed well in various downstream tasks in the field of vision, showing strong potential. [33] uses the super-pixel method to convert the classic image classification dataset into a graph structure, demonstrating competitive performance on the image recognition task. [34], [35] design graph mapping modules to learn graph representation from 2-D visual data and strengthen the relationship between objects. To solve the problem of long-range dependency capture, [36], [37] achieve the global modeling by designing global graph reasoning modules and performing prominently in the related tasks. The above work is devoted to mapping the object features into graph nodes and establishing the relationship between these objects. This mapping process has no supervision information, which leads to the node learning process and the results being unpredictable, causing unclear semantics.

2) Challenges Faced on Remote Sensing Object Detection:

a) In remote sensing scenes, objects are often closely arranged and vary in scales, which increases the difficulty of extracting object features. How to extract only the target features without mixing the noise of the neighboring targets and reducing the feature difference of objects of the same category of different scales is the key to the graph structure mapping process.
b) Remote sensing objects are generally a kind of rigid body with a high degree of symmetry, and their aspect ratios are often extreme. How to use the graph network to extract valuable features and guide the model to decode the classification and localization results is the core of the graph network module design.
c) Remote sensing objects, such as aircraft, consist of several components of the same type and different shapes. How to effectively use the features of different components and their relative position relationships to help improve fine-grained detection of aircraft is an important way to explore the potential of graph structures in the field of remote sensing.

III. Methodology

The following content will be organized into four sections. They are arranged as the problem setup, GFP, GANs, and loss function & postprocess.

A. Problem Setup

There is a kind of feature that changes slightly when the relative position, direction, and scale of a remote sensing object are changed, which we name invariable structural feature. The apparent structure of an object in the remote sensing scene is relatively stable, and the foreground targets and the

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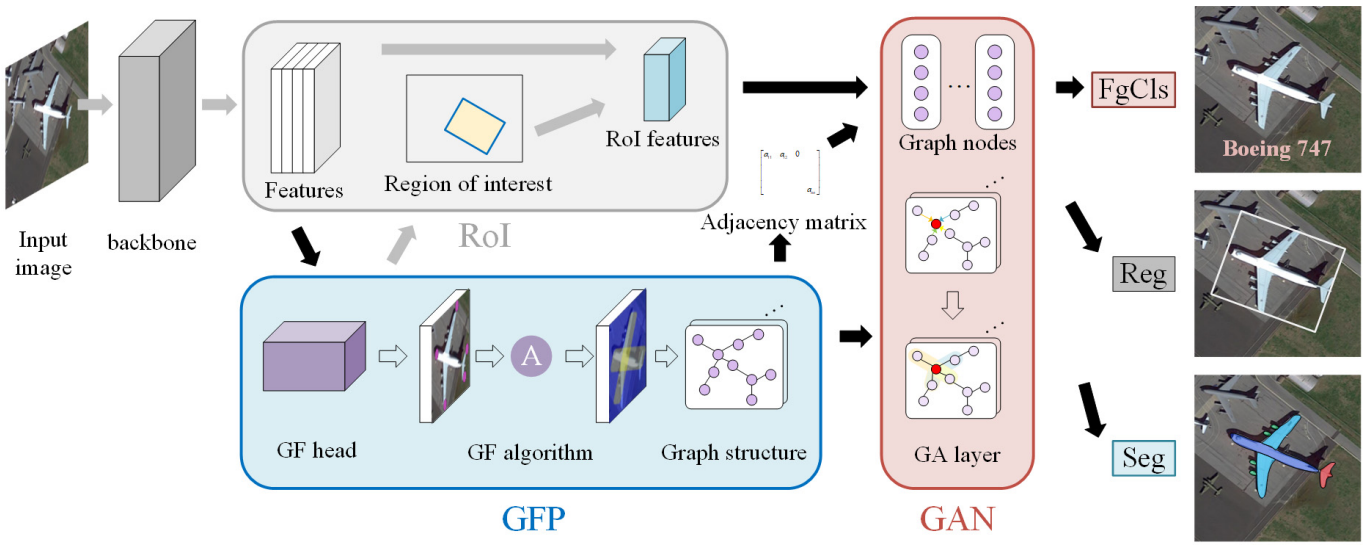


Fig. 4. Above figure shows the inference architecture of GFA-Net. The input image first passes the backbone network with the FPN to obtain the pyramid level feature map. The GFN extracts the proposals on the pyramid level feature map, and the other side processes the edge information to convert it into the graph structure representation. GAN receives the representation of the proposal features, aggregates the feature between each node, and, finally, outputs the corresponding task prediction results.

background can be distinguished intuitively. The apparent structure of a remote sensing rigid object is a kind of invariant structural feature (as shown in Fig. 3).

In order to extract the invariant structural features that can stably represent the remote sensing objects, we make our improvements based on the classic model Faster-region-based CNNs (RCNNs) [38]. The improved model is also divided into two stages. The first stage is responsible for extracting class-agnostic region proposals, and the second stage applies a classifier to distinguish the categories of region proposals and further refine them (method comparison is shown in Fig. 2).

During the first stage, the model obtains relatively stable rigid structure features by learning the contour of an object, which is used to locate the proposal regions. The specific representation method of the object contour is given as follows:

$$R_C = \{(p_{xi}, \quad p_{yi})\}_{i=1}^{N} \tag{1}$$

where R_C indicates the proposal region contour and (p_{xi}, p_{yi}) is defined as the corner coordinates in the contour.

Specifically, in the rotating object detection task, $N = 4$, which means that the contour degenerates into a rotating rectangle.

In the second stage, we further map the features inside the contour regions to the feature space represented by the graph structure

$$R_{\mathcal{G}} = (\mathcal{V}, \mathcal{E}) \tag{2}$$

where $\mathcal{V}$ represents the graph nodes and $\mathcal{E}$ can be seen as the graph edges.

After applying the graph network to complete the information interaction between nodes, we further refine the object representation according to different tasks. Here is an example

$$R_B = (d_{cx}, d_{cy}, d_h, d_w, d_\theta) \tag{3}$$

where R_B indicates the rotating bounding box represented in polar coordinate, and $(d_{cx}, d_{cy}, d_h, d_w, d_\theta)$ represent the central point offsets, anchor box scale offsets, and the angular deviation, respectively.

According to the above representations, we design a remote sensing multitask model based on graph modeling (as shown in Fig. 4). It can process three remote sensing vision tasks, namely, the remote sensing rotating object localization task, the remote sensing object recognition task, and the remote sensing component segmentation task, simultaneously. Specifically, we design a GFP to replace the traditional RPN [38]. Our GFP consists of two parts, namely, the GFN and the graph focusing algorithm (GFA). The graph focus algorithm is composed of an algorithm and a series of optional strategies. The algorithm takes the RoI feature and proposal contour obtained by GFN as input to obtain the final graph structure. Finally, we propose a GAN, which can aggregate the feature information between nodes in a graph structure so that the features between nodes of the same type can be stably expressed. We instantiate GAN by merging several graph aggregation layers (GALs) to replace the decoding process of the RoI feature, which usually utilizes the fully connected layers.

B. Graph Focusing Process

GFP consists of two parts, namely, the GFN and the graph focusing algorithm (GFA).

1) Graph Focusing Network: The traditional RPN structure only outputs the axis-aligned range of the proposal, leaving the feature misalignment problem between remote sensing objects and the relative coarse proposals. By applying the corner points representation method, GFN can not only improve the quality of the generated proposal regions but also establish the connection between the graph structure and the remote sensing object with richer texture features. We follow the same RPN network architecture and obtain a more refined representation

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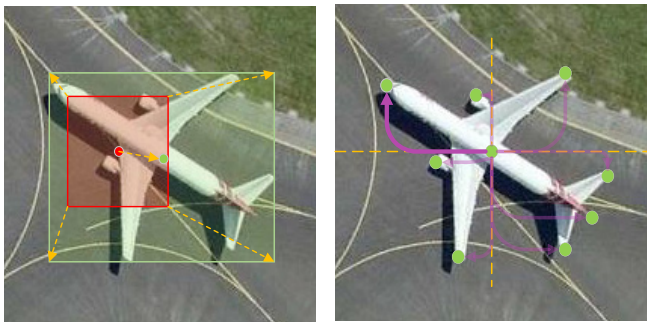


Fig. 5. Comparison of traditional RPN and our GFN. Traditional RPN outputs rough localization information of the class-agnostic proposals, and the foreground targets are based on only the center point of the object feature. Our GFN outputs the edge corners of the proposals, and the foreground targets are based on a set of predicted edge corners, including the midpoint of the objects.

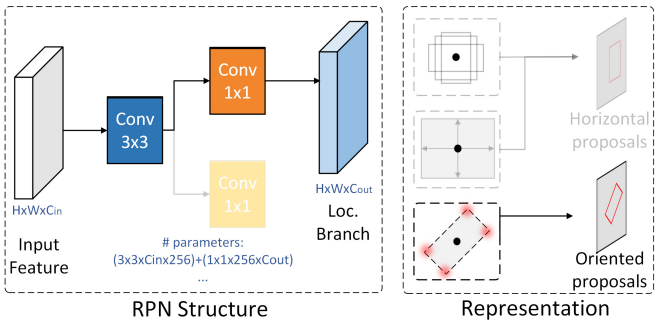


Fig. 6. We follow the same RPN architecture and design different object representation methods. As shown in the figure, compared to the traditional method of proposing a horizontal box, our representation method obtains a more refined rotating box by optimizing the measurement of the distance of the corner points distribution.

of the proposal area under the premise of nearly the same number of parameters (see Fig. 5).

Traditional RPN takes a $W \times H \times C_{\text{in}}$ feature map I ($C_{\text{in}} = 256$ usually, and batch is not considered) as input and selectively corrects the preset $\sum_{i=1}^{L} W/\Delta_i \times H/\Delta_i \times N$ anchor boxes. Here, L is the number of feature pyramid networks (FPNs) layers, Δ_i is the downsampling ratio corresponding to each FPN layer, and N means the preset number of boxes on each anchor. Each anchor box is represented by the center point coordinates, and the box length and width, which consist of a 4-D vector (c_x, c_y, d_w, d_h). The network achieves the extraction of the RoI by learning the encoding difference between the ground truth (GT) and the anchor box

$$\begin{aligned} \delta_{cx} &= (g_x - c_x)/d_w, \quad \delta_{cy} = (g_y - c_y)/d_h \\ \delta_w &= \log(g_w/d_w), \quad \delta_h = \log(g_h/d_h) \end{aligned} \tag{4}$$

where (g_x, g_y, g_w, g_h) is the GT, and the model aims to output the predicted value $\hat{\delta}_k \to \delta_k, k \in \{cx, cy, w, h\}$.

Same with fed a $W \times H \times C_{\text{in}}$ feature map, our GFN focuses a total of $\sum_{i=1}^{L} W/\Delta_i \times H/\Delta_i$ predicted point sets to the object corners as the RoIs to construct a refined object representation. The number of points set in each feature map bin is fixed and can be expressed as follows:

$$\delta_{px} = (\hat{p}_x - p_x)/\Delta_i, \delta_{py} = (\hat{p}_y - p_y)/\Delta_i \quad (i = 1, \ldots, L) \tag{5}$$

where (p_x, p_y) is the predefined corner coordinates, and $(\hat{p}_x, \hat{p}_y)$ is the predicted points' coordinates. The model strives to minimize $\delta_k \to 0, k \in \{p_x, p_y\}$.

Compared with traditional RPN, the proposed GFN has the following advantages.

a) Refinement degree differences in localizing the RoIs: In remote sensing scenes, the objects are usually rigid structures densely arranged with arbitrary orientations. Intuitively, directly learning the corners of an object can provide more reasonable proposals for the next stage compared to learning an axis-aligned area.

b) Better coupling capability with the graph network module: The corresponding features of the object corners are crucial to the classification subtasks in the next stage. The refined localization features output by GFN can be further introduced into the next stage to assist the classification branch prediction, but it is powerless for RPN.

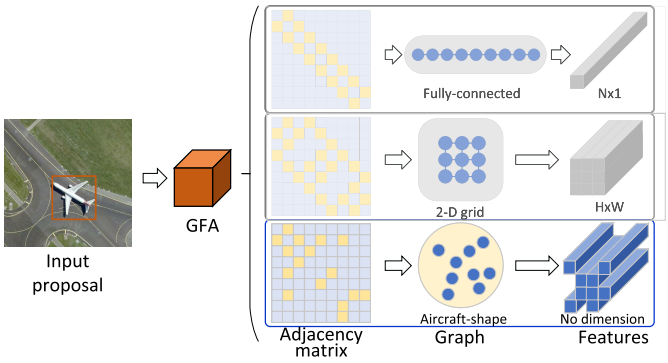


Fig. 7. Feature expression difference of fully connected layers (first row), CNNs (second row), and graph network processing (third row). It can be intuitively observed that the graph structure is more detailed in modeling the relative positional relationship of object features.

2) Graph Focusing Algorithm: As shown in Fig. 6, the high-quality proposals generated by GFN are sent to the extractor to obtain the rotating RoI features. The generated object corners will be fed into the GFA, which is responsible for focusing on the key parts of the object and is further converted into a graph structure representation.

A graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ can be applied to model the input feature X, with $\mathcal{V}$ as the nodes and $\mathcal{E}$ as the edges. Nodes $v_i \in \mathcal{V}$ are used here to represent elements x_i on the feature map X. Edges $(v_i, v_j) \in \mathcal{E}$ correspond to the connection between any two feature map elements, which can be described with an adjacency matrix $A \in \mathbb{R}^{\text{HW}} \times \text{HW}$. As shown in Fig. 7, a specific adjacency matrix can describe nodes in the state of fully connected, 2-D grids, and even more complex feature map connection structures. In other words, constructing an effective adjacency matrix can help the graph network extract object features more efficiently.

Algorithm 1 shows the pseudocode of the GFP. It inputs RoIs and corner points, and outputs the adjacency matrix used to construct the graph structure. Step 5 is implemented based on the Opencv package. Regarding the adjacency matrix

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Algorithm 1 Graph Focusing Algorithm

Input: The point set P_n containing object edges; The input RoI size (H_r, W_r); $n \ll H_r \times W_r$;

Output: The adjacency matrix A_0 describing the connection status in RoI features;

1: initialize a zero matrix $A_0 \in \mathbb{R}^{H_r W_r \times H_r W_r}$;
2: Normalize the point set offsets y to $[0, H_r/2]$, x to $[0, W_r/2]$;
3: Initialize a zero matrix $T_0 \in \mathbb{R}^{H_r \times W_r}$;
4: **for** $i \in [0, n]$ **do**
5: Calculate the minimum area rectangle mask $T_i \in \mathbb{R}^{H_r \times W_r}$ of the point subset $(p_i, p_{(i+1)\%n}, p_{(i+2)\%n}, p_{(i+3)\%n})$;
6: $T_0 = T_0 + T_i$;
7: **end for**
8: Normalize $t_{ij} \in T_0$ to $[0, 1]$;
9: **for** $t_{ij} \in T_0$ **do**
10: Assign t_{ij} to relevant elements of matrix A_0 based on selective strategies;
11: **end for**
12: **return** A_0;

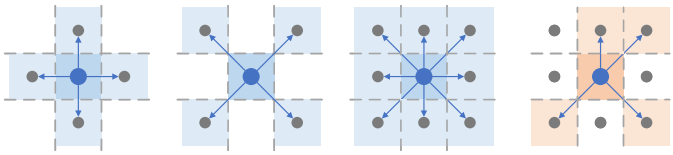


Fig. 8. Four connection strategies between nodes for selection. They are strategies A, B, C, and D (from left to right). Related ablation experiments and strategy selection are in the experimental section. Blue represents the manual design method, and red represents the model adaptive method.

A_0 assignment strategy in step 10, four options are provided (as shown in Fig. 8), which will be further tested in subsequent ablation experiments.

The axis-aligned RoIs' output from RPN contains not only the features of the object itself but also complex background features. The lack of spatial information in the fully connected layer may interfere with the distinction between the above two, thereby affecting the recognition and localization results. The graph focusing algorithm takes the edge information of an object as input and obtains valuable foreground features for distinguishing from the relatively disturbing background features. A coarse mask for the object edges is generated by the designed graph focusing algorithm. As shown in Fig. 7, it aims to provide a more representative nodes connection form for the next stage of GAN.

C. Graph Aggregation Network

As mentioned in Section III-B2, GFA expresses the connection state of the nodes that have completed spatial focus in the form of the adjacency matrix $A \in \mathbb{R}^{N \times N}$. It serves as an important weighting factor of the GAL to provide the focusing area of the features that only contain pure foreground information. Next, we will introduce the GAL which aims to further aggregate the features of the focusing area.

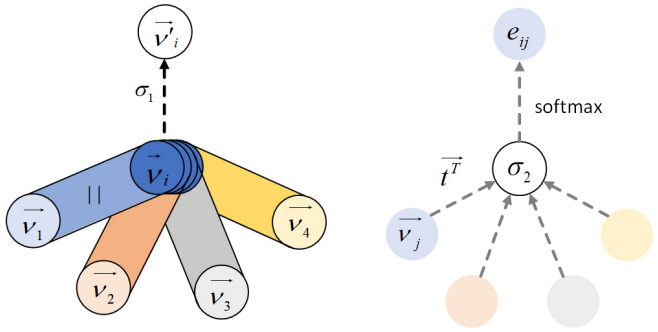


Fig. 9. Figure on the left shows the update process of the node feature $\vec{v}_i$, and the figure on the right shows the calculation process of the fusion factor e_{ij}.

We start from a single GAL. Its input contains a adjacency matrix A and a set of node features $V = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_N\}, \vec{v}_i \in \mathbb{R}^C$, where N represents the node amount and C is the channel of each node feature. After processed by a GAL, the same number of nodes' features with different channels will be produced $V' = \{\vec{v}'_1, \vec{v}'_2, \ldots, \vec{v}'_N\}, \vec{v}'_i \in \mathbb{R}^{C'}$.

We first define a learnable shared linear transformation matrix $W \in \mathbb{R}^{C' \times 2C}$, which is utilized to extract features on each node. For each one, we obtain the set of all its one-hop neighbor nodes $\mathcal{N}_i$ according to the derived adjacency matrix A. The feature updating formula of each node is expressed as follows:

$$\vec{v}'_i = \sigma_1 \left(\sum_{j \in \mathcal{N}_i} e_{ij} W \left[\vec{v}_j || \vec{v}_i \right] \right). \tag{6}$$

Among them, σ_1 represents the activation function (usually a nonlinear operation). $||$ means concatenating two vectors. e_{ij} represents the normalized fusion factor between (v_i, v_j), which is used to measure the fusion proportion of each one-hop neighbor node. Its calculation method is listed as follows:

$$e_{ij} = \frac{\exp\left(\sigma_2\left(\vec{t}^T \vec{v}_j\right)\right)}{\sum_{k \in \mathcal{N}_i} \exp\left(\sigma_2\left(\vec{t}^T \vec{v}_k\right)\right)} \tag{7}$$

σ_2 also represents the nonlinear activation function. Here, we introduce a high-dimensional mapping weight vector $\vec{t} \in \mathbb{R}^C$, which is responsible for extracting the fusion importance of the features of each one-hop neighbor node. Here, $\cdot^T$ represents the transpose operation. The whole process can be described in Fig. 9.

We also introduce a multihead mechanism to stabilize the divergence caused by a single fusion factor. Referring to [39], the final layer output will be obtained by concatenating M groups of $\vec{v}'_i$ independently calculated by 6. The multihead graph fusion layer calculation process can be expressed as follows:

$$\vec{v}'_i = \Big\Vert_{m=1}^{M} \sigma_1 \left(\sum_{j \in \mathcal{N}_i} e_{ij}^m W^m [\vec{v}_j || \vec{v}_i] \right) \tag{8}$$

where e_{ij}^m and W^m both represent the fusion factor and the linear transformation matrix under the mth group head mechanism. Note that the output dimension of W^m is $\mathbb{R}^{C'/M \times 2C}$,

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maintaining the same dimension with 6 after concatenation. At the same time, we provide a variant of the multihead mechanism to adapt to different structures

$$\vec{v}_i' = \sigma_1\left(\frac{1}{M}\sum_{m=1}^{M}\sum_{j\in\mathcal{N}_i} e_{ij}^m W^m\left[\vec{v}_j \middle|\middle| \vec{v}_i\right]\right). \tag{9}$$

1) Instantiation: We design GAN to instantiate the classifier module corresponding to the second stage of the two-stage model. Specifically, GAN takes the RoI feature $X \in \mathbb{R}^{W\times H\times C}$ and the corresponding adjacency matrix A as input. In GFA, we adopt strategy $C\&D$ for the assignment of elements of the adjacency matrix A. At the same time, we stack a two-layer GAN. The first layer adopts the multihead mechanism of $M = 8$ and outputs the feature channel $C = 128$ after concatenation operation. Among them, σ_1 applies exponential linear unit (ELU) activation [40], and σ_2 applies LeakyReLU activation ($\alpha = 0.2$ for negative slope) [41]. The last layer has two branches that are responsible for the multilabel classification and object localization regression. The classification branch adopts the graph fusion layer following the averaging mechanism and outputs the feature channel of the class numbers. The localization branch derives the object localization through a simple linear transformation.

D. Loss Function & Postprocess

1) Loss Function: The loss function contains two stages. GFN applies a mixture of Intersection over Union (IoU) loss and chamfer distance (CD) loss to guide the learning of object edge point sets. IoU is used to describe the intersection ratio between two sets of axis-aligned boxes, and CD can be further measured to describe the difference in point set distribution between two contours. The above two complement each other. Cross-entropy loss is adopted for the learning of class-agnostic region proposal subtask

$$L_{\text{stage1}} = \alpha(L_{\text{IoU}} + L_{\text{CD}}) + L_{\text{CE}} \tag{10}$$

where $\alpha = 0.5$. The GAN in the second stage utilizes cross-entropy loss and smooth L1 loss in classification and localization, respectively, which follows the implementation in Faster-RCNN [38] for rotating object detection task (Faster-RCNN-R)

$$L_{\text{stage2}} = \beta L_{\text{Smooth}L1} + L_{\text{CE}} \tag{11}$$

where $\beta = 1$. The total loss function is expressed as

$$L_{\text{total}} = \gamma_1 L_{\text{stage1}} + \gamma_2 L_{\text{stage2}}. \tag{12}$$

We set $\gamma_1 = \gamma_2 = 1$.

2) Post Process: The postprocess is consistent with Faster-RCNN-R. We apply the rotating nonmaximum suppression (NMS) to process the detected rotating bounding boxes to derive the final detection results.

IV. Experiments

A. Datasets and Settings

We apply the designed model to two downstream tasks in the remote sensing vision domain. For the task of rotating object detection, we select three classic and representative public datasets: DOTA [42], HRSC2016 [43], and FAIR1M [19]. For the component segmentation task, we propose a self-made dataset named ACSD. To the best of our knowledge, it is the first remote sensing dataset annotated for the extraction of aircraft components.

1) *DOTA [42]:* For the oriented object detection task, it consists of 2806 images covering from 800 × 800 to 4000 × 4000 pixels. A total number of 188 212 instances are annotated in 15 common categories, which includes plane (PL), basketball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC).
2) *HRSC2016 [43]:* With an average aspect ratio of 5:1, its instances are annotated in the way of an oriented bounding box (OBB). The image resolutions cover from 0.4 to 2 m, and sizes are between 300 × 300 and 1500 × 900 pixels. We train our model on the training set and the validation set, and test the results on the test set. In terms of data augmentation, we rescale the images to (800, 1333) and apply random horizontal flips.
3) *FAIR1M [19]:* As a large-scale fine-grained remote sensing dataset, FAIR1M contains more than 1 million instances and more than 15 000 images with a resolution of 0.3–0.8 m from different platforms. The whole objects are annotated concerning five categories and 37 subcategories by OBBs. We obtain part of the dataset and divide the training set and the test set into five categories according to the range of 4:1. Finally, we verify the effectiveness of our method on fine-grained recognition tasks in these five category subdatasets, respectively.
4) *ACSD:* The proposed dataset adopts two labeling methods, including component segmentation and the OBB for aircraft. For segmentation labeling, each aircraft is divided into four categories according to its components: airframe, wing, engine, and tail. For OBB labeling, six categories have been labeled according to their types. Containing 6598 instances, the images of the ACSD dataset that we collected are derived from the DOTA dataset and the FAIR1M aircraft dataset. They both are collected from multiple sensors (e.g., Gaofen) and platforms (e.g., Google Earth) with multiple resolutions. The image sizes in ACSD cover from 384 × 384 to 640 × 640 pixels, with resolutions ranging from 0.4 to 2 m. The dataset distribution and visualization are shown in Fig. 10.

During experiments, Resnet101 with FPN is applied for model comparison, and Resnet50 with FPN is used for ablation studies. We completely follow the mmdetection [44] training schedules. All models are trained in 12 epochs on the DOTA dataset and in 36 epochs on HRSC2016 and ACSD. The learning rate is initialized to 0.01 with a stochastic gradient descent (SGD) optimizer, and after each preset decay step, it will be divided by 10. We set the weight decay and momen-

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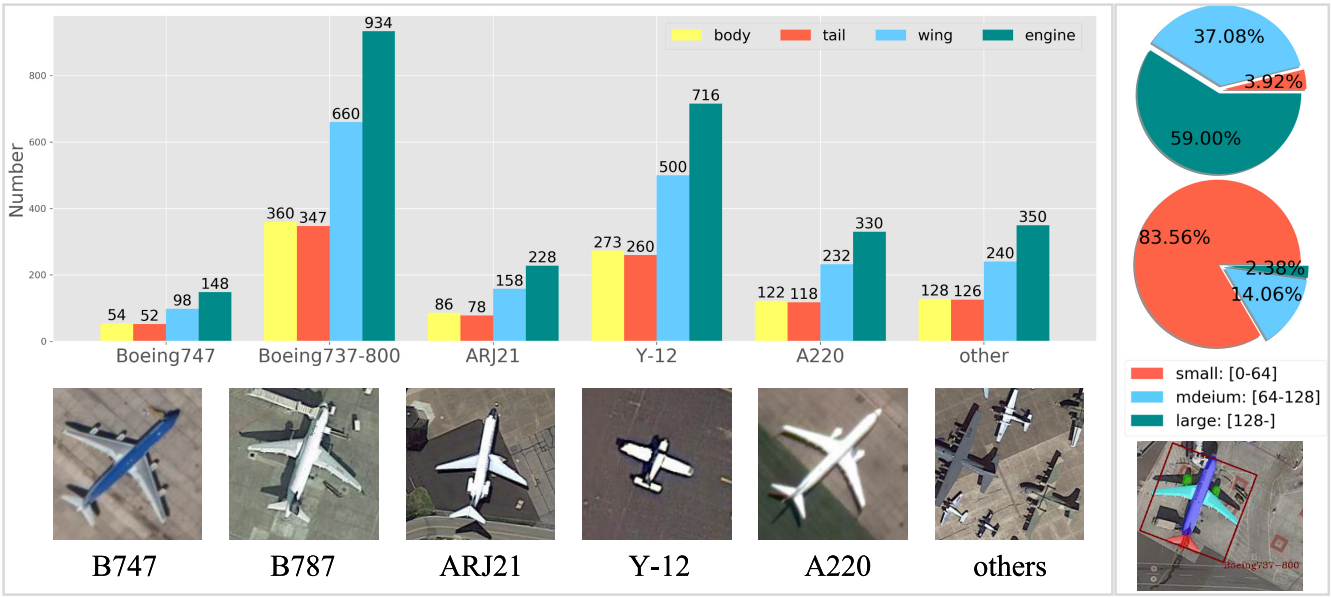


Fig. 10. Statistics about ACSD. The numbers of components in each category (above) and visualization instances (below) are described in the left part. From top to bottom in the right part is category-level and component-level object scale pie charts and the ACSD labeling method (rotating box and component segmentation).

tum to 10^{-4} and 0.9, respectively. A warmup for learning rate is applied during the first 500 iterations. We utilize four P100 graphics processing unit (GPU) with a two batch size for each while training, and one single P100 GPU for testing. The sliding window strategy is adopted to crop the large remote sensing images to send them into the models.

B. Evaluation Metrics

Average precision (AP) and average recall (AR) are adopted to evaluate the performance of the models on the rotating object detection task and the instance segmentation task. The rotating bounding box or the predicted mask that overlaps with the GT may be regarded as a candidate. After obtaining the candidates, we further calculate the intersection over union (IoU) between each pair. The objects that meet the preset conditions can be considered as correctly positioned [42], [45], [46].

1) Precision and Recall: IoU refers to calculating the ratio of the intersection and union between the predicted object (box or mask) and the GT. True positive (TP) means a predicted object whose IoU with the unmatched GT is larger than the threshold. False positive (FP) is just the opposite. After matching all the predicted objects, the remained unmatched GT is called false negative (FN).

The following equations can be used to calculate the precision and recall:

$$\text{precision} = \frac{\text{TP}_s}{\text{TP}_s + \text{FP}_s} \tag{13}$$

$$\text{recall} = \frac{\text{TP}_s}{\text{TP}_s + \text{FN}_s} \tag{14}$$

where $\cdot_s$ represents the number of all instances obtained on a certain dataset.

2) AP: IoU $>$ 0.5 [45] is adopted in our experiments while calculating precision and recall. We also choose IoU $>$ 0.7 for exploring the potential performance of our model. Finally, AP can be derived by calculating the area between the precision–recall (PR) curve and its coordinate axis. If a multicategories task is needed, we define the mean AP of each category as mAP.

3) AR: We follow the common objects in context (COCO) measurement [45] to calculate AR, and the equation is given as follows:

$$\begin{aligned}\text{AR} &= 2\int_{0.5}^{1} \text{recall}(\text{IoU})d(\text{IoU}) \\ &= \frac{2}{n}\sum_{i=1}^{n} \max(\text{IoU}(gt_i - 0.5), 0)\end{aligned} \tag{15}$$

where n is the number of overlaps between all GTs in each image and the predicted objects closest to them. In the experiments, we take $n = 1000$ and denote as AR_{1000}.

C. Comparisons With Sate-of-the-Art Methods

1) Comparisons on Rotating Object Detection Task: We compare the performance of our GFA-Net with the classic models that are based on structural modular design and object representation modeling. They are under the same experimental environment on the listed datasets. A modified FasterRCNN-R [38] detector for the rotating object detection task is provided as the baseline model.

a) Experiment on DOTA: On the DOTA dataset, we list the proposed GFA-Net with other SOTA detectors under two evaluation metrics. They are mAP under the circumstance of IoU $=$ 0.5 (standard) and IoU $=$ 0.7 (more strict for localization), consistent with other models. We choose Resnet-101 with FPN as the backbone architecture of the proposed

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TABLE I

SOTA COMPARISONS ON DOTA. ALL MODELS USE THE SAME BACKBONE RESNET-101 [5]. "FPS" MEANS FRAMES PER SECOND. "-R" MEANS VERSION FOR THE ROTATING OBJECT DETECTION TASK

Models	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	FPS	mAP
																	IoU(0.5)
RetinaNet-R [47]	81.1	53.2	33.6	45.7	48.5	55.7	75.9	90.0	42.6	64.3	44.4	59.3	54.3	61.7	49.1	9.2	57.3
R-DFPN [48]	80.9	65.8	33.8	58.9	55.8	50.9	54.8	90.3	66.3	68.7	48.7	51.8	55.1	51.3	35.9		57.9
R^2CNN [26]	88.5	71.2	31.7	59.3	51.9	56.2	57.3	90.8	72.8	67.4	56.7	52.8	53.1	51.9	53.6		60.7
RRPN [22]	80.9	65.8	35.3	67.4	59.9	50.9	55.8	90.7	66.9	72.4	55.1	52.2	55.1	53.4	48.2		61.0
Yang et al. [49]	81.3	71.4	36.5	67.4	61.2	50.9	56.6	90.7	68.1	72.4	55.1	55.6	62.4	53.4	51.5		62.3
FR-R [38]	88.0	77.0	36.7	72.5	70.2	61.8	75.8	90.1	73.8	85.0	56.6	62.6	53.3	59.5	41.9	6.2	67.0
ICN [50]	81.4	74.3	47.7	70.3	64.9	67.8	70.0	90.8	79.1	78.2	53.6	62.9	67.0	64.2	50.2		68.2
RoITrans. [16]	88.6	78.5	43.4	75.9	68.8	73.7	83.6	90.7	77.3	81.5	58.4	53.5	62.8	58.9	47.7	5.9	69.6
CAD-Net [24]	87.8	82.4	49.4	73.5	71.1	63.5	76.6	**90.9**	79.2	73.3	48.4	60.9	62.0	67.0	62.2		69.9
R^3Det [20]	89.5	82.0	48.5	62.5	70.5	74.3	77.5	90.8	81.4	83.5	62.0	59.8	65.4	67.5	60.1		71.7
SCRDet [23]	90.0	80.7	52.1	68.4	68.4	60.3	72.4	**90.9**	**87.9**	**86.9**	65.0	66.7	66.3	68.2	65.2		72.6
O^2-DNet [51]	89.3	83.3	50.1	72.1	71.1	75.6	78.7	**90.9**	79.9	82.9	60.2	60.0	64.6	68.9	**65.7**		72.8
FR-Est [30]	89.6	81.2	50.4	70.2	73.5	78.0	86.4	90.8	84.1	83.6	60.6	66.6	70.6	66.7	60.6		74.2
AOPDet [29]	90.2	77.9	47.5	74.3	73.6	78.7	**89.5**	**90.9**	66.4	85.7	57.1	71.9	75.6	72.8	60.9	8.8	74.3
Gliding Vertix [27]	89.6	**85.0**	52.3	**77.3**	73.0	73.1	86.8	90.7	79.0	86.8	59.6	70.9	72.9	70.9	57.3	10.0	75.0
S^2A-Net [21]	88.7	81.4	**54.3**	69.8	**78.0**	**80.5**	88.0	90.7	84.8	86.2	65.0	65.8	**76.2**	**73.4**	58.9	**12.7**	**76.1**
GFA-Net	**90.3**	74.3	51.6	72.1	72.5	76.8	**89.5**	**90.9**	65.9	86.5	**69.5**	**75.0**	73.4	71.9	64.3	11.2	75.0
																	IoU(0.7)
FR-R [38]	80.4	39.4	15.1	53.0	24.5	49.2	42.4	89.8	50.7	62.3	59.4	43.5	29.1	22.3	25.0	6.2	45.7
RoITrans. [16]	80.6	50.4	16.9	57.4	24.3	52.1	47.2	90.4	51.9	66.1	63.4	50.9	40.9	24.4	27.1	5.9	49.6
AOPDet [29]	81.0	49.3	19.6	61.1	24.8	60.1	50.3	90.7	64.7	64.9	72.6	52.6	50.5	26.7	26.1	8.8	53.0
GFA-Net	**88.9**	58.9	23.5	63.3	38.5	63.3	73.2	90.7	63.7	74.8	62.1	59.3	41.5	31.0	32.7	11.2	57.7

TABLE II

COMPARISONS WITH THE SOTA MODELS ON HRSC2016

method	RRD [52]	RoI Trans.[16]	Gliding Vertix [27]	AOPDet [29]	R^3det [20]	S^2A-Net [21]	GFA-Net
mAP	84.3	86.2	88.2	88.5	89.3	90.2	90.5

GFA-Net for consistency with other detectors [16], [23], [24], [30], [47]. Only GFA-Net applies the graph modeling technique to process the extracted features, while others remain processing features in the polar coordinate space or the Euclidean space.

The results are listed in Table I. Under mAP(IoU = 0.5), GFA-Net achieves 75.0% and 11.2 FPS. It surpasses all listed detectors except for S^2A-Net, deriving a satisfactory speed–accuracy tradeoff. In the 15 categories' comparison, there are five in which GFA-Net reaches the SOTA level. Specifically, compared with the baseline Faster-RCNN Rotated (FR-R), GFA-Net has achieved a performance improvement of 8% mAP and 2.4 FPS. The only difference lies in the modeling method of the extracted features, while GFA-Net adopts the graph structure modeling. Under mAP(IoU = 0.7), which is more strict for localization, GFA-Net surpasses all the fashion models listed, demonstrating a powerful localization capability. The visualization results in Fig. 11 also demonstrate the better localization and classification capabilities of the proposed method than that of the baseline model.

Our GFA-Net has 1.1 mAP and 1.5 FPS reduction compared with S^2A-Net. The possible reasons are given as follows. The DOTA dataset contains categories of rigid objects (e.g., aircraft, vehicles, and ships) and some categories of irregular shapes (e.g., bridges, ports, and swimming pools). Our proposed GFA-Net models rigid-body objects. The invariant structural features of irregular objects in the dataset are relatively more difficult to extract, so the overall performance is not as well as S2A-Net. S2A-Net is dedicated to solving the feature misalignment problem between the anchor boxes and axis-aligned convolutional features; there is no requirement for the object structure.

b) Experiment on HRSC2016: The HRSC2016 dataset contains ships with arbitrary directions, extreme aspect ratios, and various scales. Anchor-based modeling methods depend on calculating a set of predefined anchor boxes to adapt to it. The model performance strongly depends on the settings of hyperparameters such as the shape and size of the preset anchor boxes. They may not capture the global information of large-scale ships. The proposed GFA-Net first applies corner

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TABLE III

COMPARISONS WITH THE SOTA MODELS ON FAIR1M. EACH METHOD APPLIES TWO LEARNING STRATEGIES: 12 EPOCHS (LEFT COLUMN OF VERTICAL LINE) AND 36 EPOCHS (RIGHT COLUMN OF VERTICAL LINE)

Category	F. Category	RetinaNet [47]	FCOS-R [53]	Faster-RCNN [38]	RoI T.[16]	Gliding V. [27]	GFA-Net
Airplane	Boeing737	25.7|38.1	28.9|38.4	35.5|28.9	31.4|37.7	35.0|35.8	34.2|38.5
	Boeing747	47.2|71.4	77.6|83.6	58.8|77.6	70.1|81.0	65.4|83.9	76.4|85.6
	Boeing777	13.9|13.7	17.6|19.7	12.0|17.6	14.8|19.6	18.6|17.7	16.5|15.5
	Boeing787	31.7|30.6	52.5|53.6	47.9|52.5	53.3|59.5	47.6|57.3	53.0|60.7
	A220	40.3|53.1	56.4|55.5	50.9|56.4	55.0|45.9	51.5|44.1	53.4|51.4
	A321	52.7|68.4	62.5|68.8	70.0|62.5	74.3|72.2	70.1|67.3	73.8|66.9
	A330	40.4|42.3	71.7|75.4	76.0|71.7	74.8|76.4	70.1|77.9	73.5|70.6
	A350	34.6|59.2	63.8|73.4	66.9|63.8	66.6|73.6	65.1|80.3	73.9|80.8
	other airplane	61.8|71.0	65.1|63.7	71.6|65.1	70.4|70.4	66.5|64.6	67.7|64.1
	mAP	38.7|49.8	49.8|59.1	54.4|55.1	56.7|**59.6**	54.4|58.8	**58.0**|59.3
Ship	Passenger Ship	1.7| 6.7	14.2|31.7	13.9|35.8	23.9|52.1	13.8|46.4	29.6|54.0
	Motorboat	20.7|30.7	22.7|40.9	44.5|55.9	55.9|64.3	52.5|57.1	60.9|66.4
	Fishing Boat	18.3|36.3	26.8|42.6	30.6|42.5	49.9|60.0	34.1|50.0	52.9|66.2
	Tugboat	31.0|60.0	43.8|69.4	49.6|71.3	64.6|75.7	53.7|73.0	68.3|75.5
	Engi. Ship	35.7|63.0	47.3|68.5	39.7|64.7	62.4|75.8	52.1|72.9	68.1|76.9
	LC Ship	26.8|64.7	65.9|81.9	47.0|67.4	81.4|88.2	68.4|83.7	82.7|87.6
	DC Ship	36.6|66.0	59.9|80.7	55.3|63.2	82.4|87.6	71.2|76.0	84.3|88.4
	Warship	2.1|49.0	47.8|67.5	25.9|56.3	68.4|76.7	55.3|66.1	71.7|77.8
	Other ship	12.7|17.8	5.6|24.9	9.0|36.2	22.6|43.9	13.7|41.8	24.6|45.9
	mAP	20.6|43.8	37.1|56.5	35.1|54.8	56.8|69.4	46.1|63.0	**60.3**|**71.0**
Vehicle	Trailer	0.0| 1.0	7.8|31.2	7.3|23.1	8.5|43.6	15.4|39.1	16.4|37.5
	Small Car	40.6|54.8	60.7|68.3	59.9|60.7	69.2|64.8	60.1|60.7	70.1|65.6
	other vehicle	0.0| 9.8	3.2|15.6	10.5|23.0	11.2|30.3	11.8|26.7	10.8|22.8
	Bus	3.6|10.9	29.0|52.4	30.2|51.8	42.6|62.0	39.1|57.1	46.0|62.1
	Cargo Truck	25.4|40.5	63.7|71.3	54.3|58.4	66.0|72.8	66.7|71.6	70.0|74.9
	Truck Tractor	0.0| 1.0	3.2|26.6	1.0|26.9	4.3|36.7	2.3|27.9	2.2|26.7
	Tractor	0.0|10.9	15.4|34.6	13.8|49.4	24.3|64.6	22.6|56.7	25.3|65.2
	Van	37.7|52.0	59.0|67.2	60.7|61.7	69.5|65.4	61.3|61.5	70.5|66.1
	Dump Truck	13.6|44.3	45.4|60.7	62.6|63.9	66.3|71.1	62.9|64.1	67.3|71.6
	Excavator	0.0| 9.2	27.5|51.5	26.8|48.3	33.1|55.9	29.7|50.3	36.5|54.1
	mAP	12.1|23.4	31.5|47.9	32.7|46.7	39.5|**56.7**	37.2|51.6	**41.5**|54.7

points regression to find the corners of the object, eliminating the manual process of predefining the scale of the anchor boxes. The next graph network processing process realizes the interaction of global information, helping the network to better recognize objects of different scales.

The results in Table II quantify the performance of GFA-Net and other models, indicating the effectiveness of the proposed method. The visualization results in Fig. 12 also show our model's ability to locate each target in different scenarios.

c) Experiment on FAIR1M: We conduct experiments on three main parts of the large-scale fine-grained FAIR1M dataset, which are airplane, vehicle, and ship, respectively. Each category contains more than nine fine-grained categories, whose apparent structure is flexible and slightly deformed. RetinaNet [47] and Faster-RCNN [38] are baseline models for one- and two-stage methods, respectively. Fully convolutional one-stage object detector (FCOS)-R [53] is based on single-stage method architecture and replaces the anchor box mechanism with a representation method of distance measurement. RoI transformer [16] extracts invariant structural features by designing the structural modules, while gliding vertex [27] attempts to achieve it by using the geometric representation of a quadrilateral. The proposed GFA-Net models

TABLE IV

COMPARISONS ON ACSD. ALL MODELS UTILIZE THE SAME BACKBONE RESNET-50 [5] WITH FPN. "ROD" MEANS THE ROTATING OBJECT DETECTION TASK, WHILE "ACS" MEANS THE AIRCRAFT COMPONENT SEGMENTATION TASK. "-" MEANS NO RESULTS ON THE CORRESPONDING TASK

Model	B747	B787	ARJ21	Y-12	A220	others	mAP	body	wing	tail	engine	mAP
ROD												
FR-OBB	53	34	46	49	56	76	65.1	-	-	-	-	-
RoI trans.	77	37	73	70	62	76	72.3	-	-	-	-	-
ACS												
Mask-RCNN								74.3	82.5	73.8	45.3	69.0
MT												
baseline							73.8	75.1	82.7	72.6	44.1	68.6
GFA-Net							75.8	75.8	83.6	75.2	48.1	70.7

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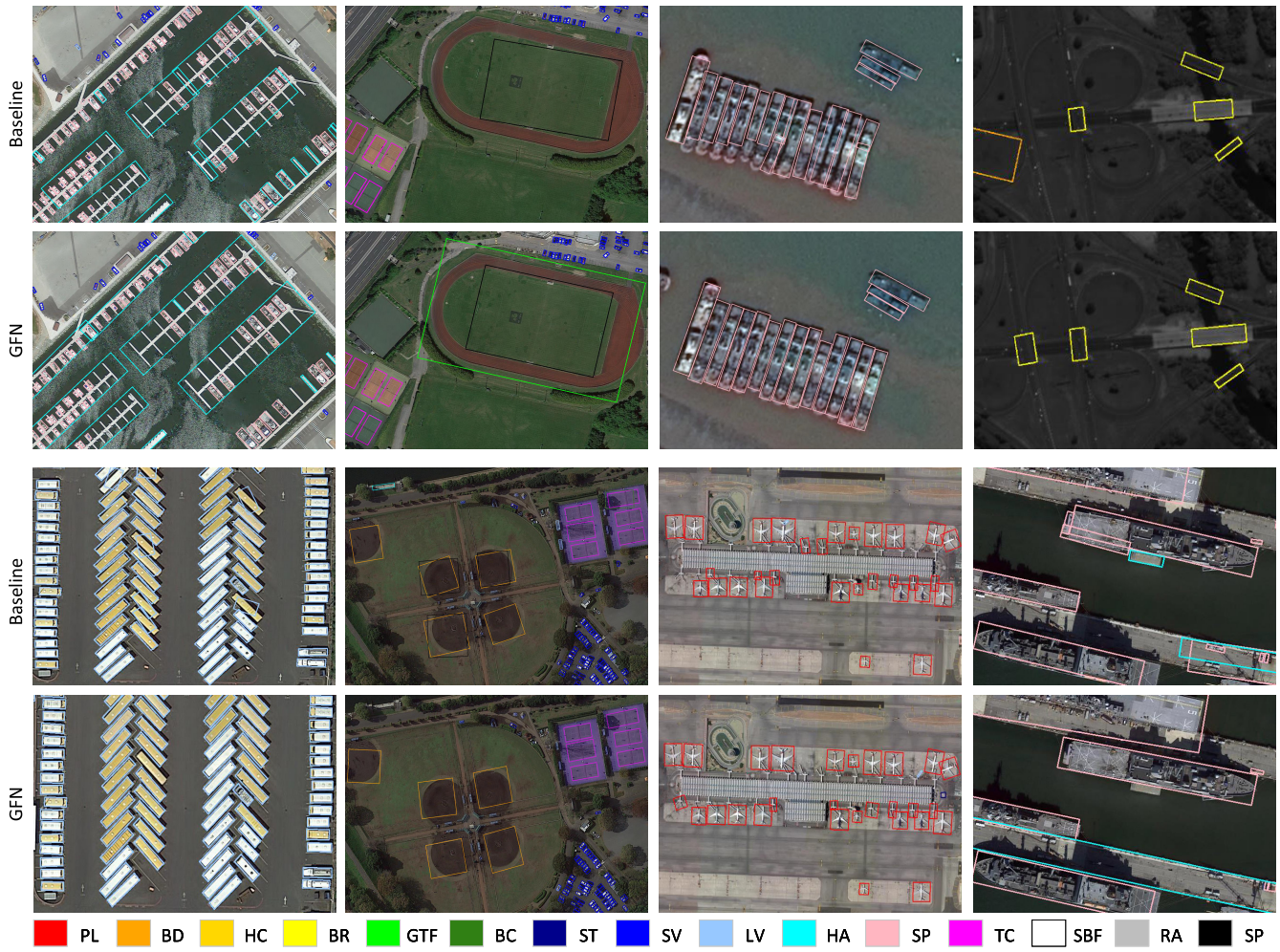


Fig. 11. Visualization comparison between the baseline method and GFA-Net on the DOTA dataset. Notice that rows 1 and 3 correspond to the baseline model, and rows 2 and 4 correspond to GFA-Net.

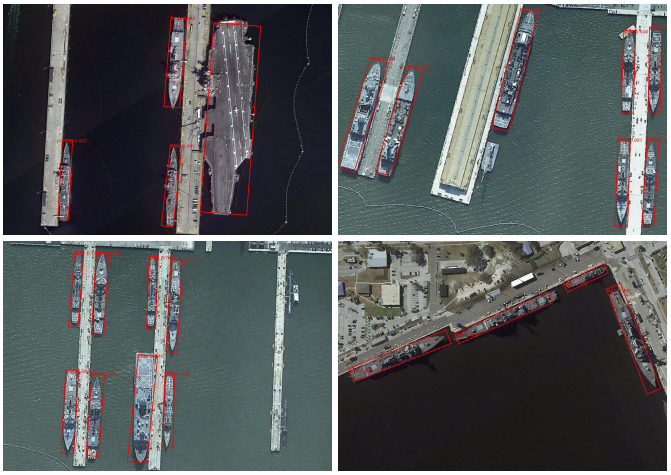


Fig. 12. Visualization results of the proposed GFA-Net on the HRSC2016 test set.

the differences in tiny deformations of objects through graph structures to obtain invariant structural features that are more differentiated and robust.

We conduct experiments on the above methods with $1\times$ and $3\times$ learning schedules. Based on the data results in Table III and the visualization results in Fig. 13, the following three conclusions can be drawn: 1) on three fine-grained categories datasets, GFA-Net surpasses the existing baseline model and the comparison model, reaching the level of SOTA; 2) our GFA-Net derives a better model convergence efficiency under

TABLE V

COMPARISONS OF GFN FOR FINE-TUNING ON DIFFERENT HYPERPARAMETERS. THE EXPERIMENTAL RESULTS ARE CONDUCTED ON THE DOTA DATASET. "POS." MEANS THE POSITION WHERE BACKBONE LAYERS CONNECT FPN, "SL" STANDS FOR THE NUMBER OF STACKED LAYERS, AND "PAR." MEANS THE PARAMETERS. BASELINE REFERS TO THE TRADITIONAL RPN STRUCTURE

Method	Pos.	Cls Loss	Loc Loss	SL	Par.	AR_{1000}
baseline	P_{2-6}	CE	Smooth L1	1	0.6	45.7
GFN	P_{3-7}	CE	Smooth L1	1	0.6	46.0(+0.3)
	P_{2-6}					46.6(+0.6)
GFN	P_{2-6}	focal CE	Smooth L1	1	0.6	46.9(+1.2)
						46.6(-0.3)
GFN	P_{2-6}	CE	Smooth L1 IoU+CD	1	0.6	46.6(+0.9)
						48.4(+1.8)
GFN	P_{2-6}	CE	IoU+CD	1	0.6	48.4(+2.7)
				3	1.7	**49.3**(+0.9)

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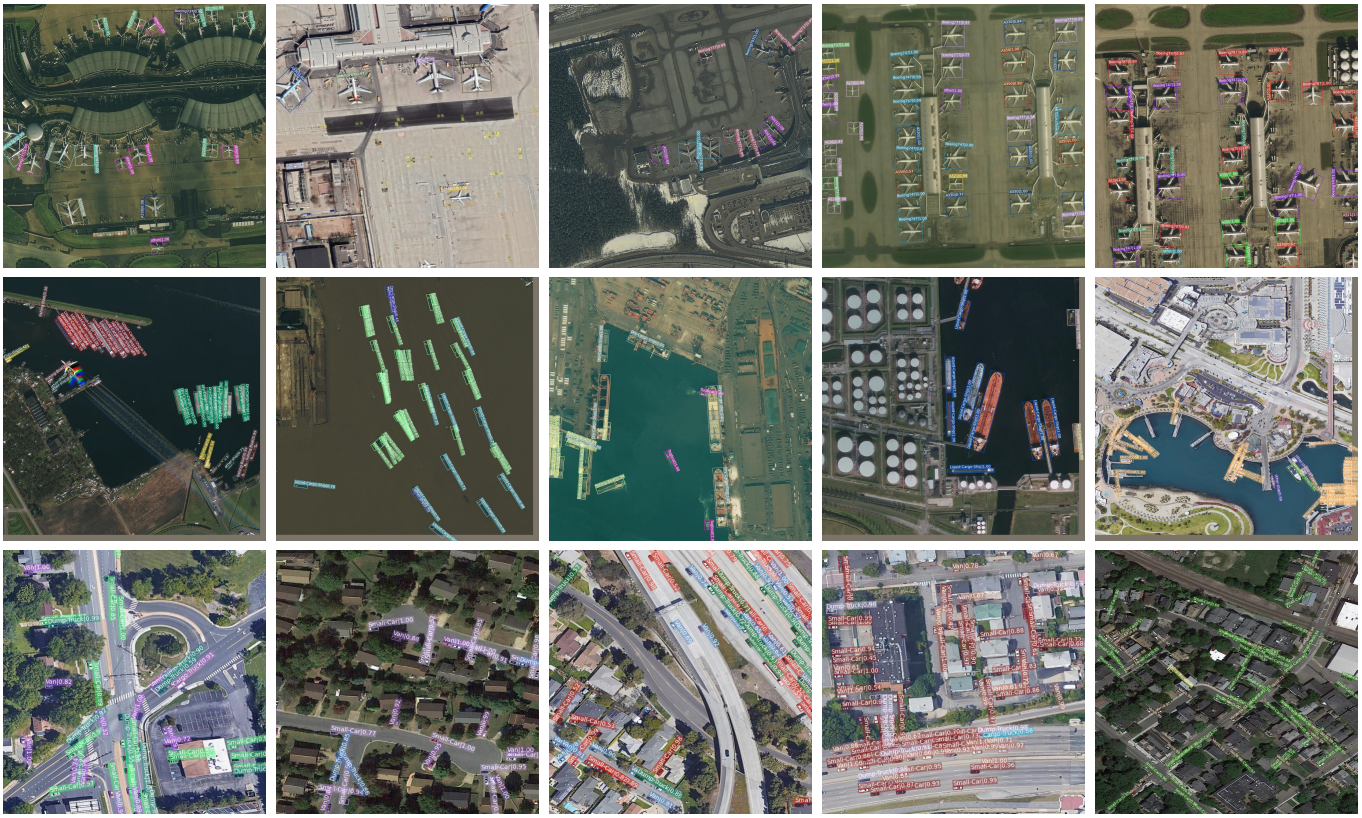


Fig. 13. Visualization results of the proposed GFA-Net on the FAIR1M dataset. Notice that rows 1–3 correspond to the airplane categories, ship categories, and vehicle categories, respectively.

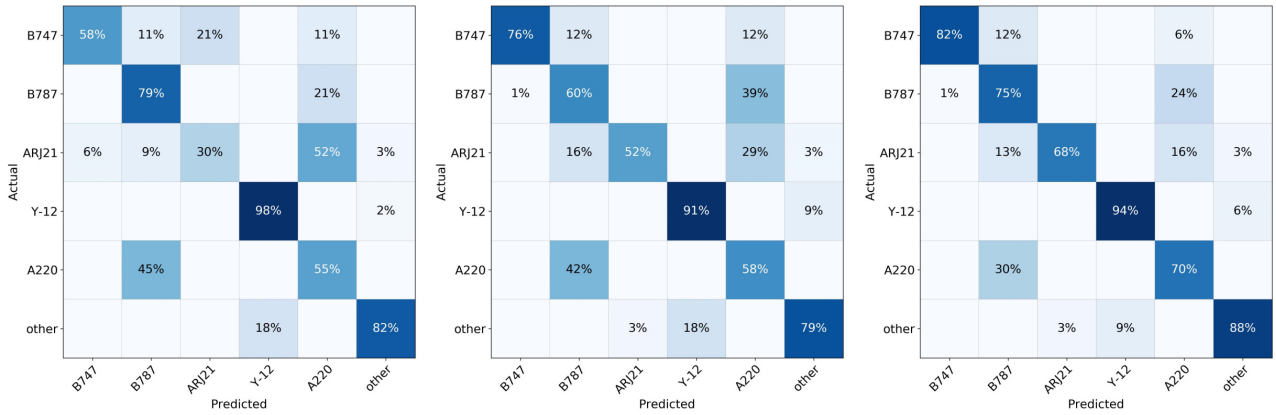


Fig. 14. Confusion matrix of the three methods on different aircraft categories. They are Faster-RCNN (OBB), baseline, and GFA-Net (from left to right).

a relatively short learning schedule (1×), and the performance is closer to the 3× result; and 3) in some difficult-to-divide fine-grained categories (B737 versus A220, LC ship versus dc ship, trailer, and tractor), GFA-Net has significantly improved performance compared to the baseline model.

Our GFA-Net outperforms on both 1× and 3× experimental results. We believe that this is due to the relatively more accurate and efficient extraction ability of graph modeling for invariant structural features of objects. It not only helps the model to converge faster but also improves the localization ability of objects. RoI transformer [16] also performs well on several experimental results, mainly due to the achievement of its rotating RoI extractor. It helps the RoI transformer to utilize the idea of cascade and step-by-step refinement to complete the extraction process of invariant structural features of objects. Other models ignore the feature extraction of object edge details, so they lack the ability to extract invariant structural features, resulting in poor performance.

2) Comparisons on Component Segmentation Task: We perform the experimental analysis of three remote sensing vision tasks on ACSD, namely, rotating object detection, component segmentation, and multitask processing.

a) Experiment on ACSD: We choose several SOTA models in their respective tasks for experimental comparison. All

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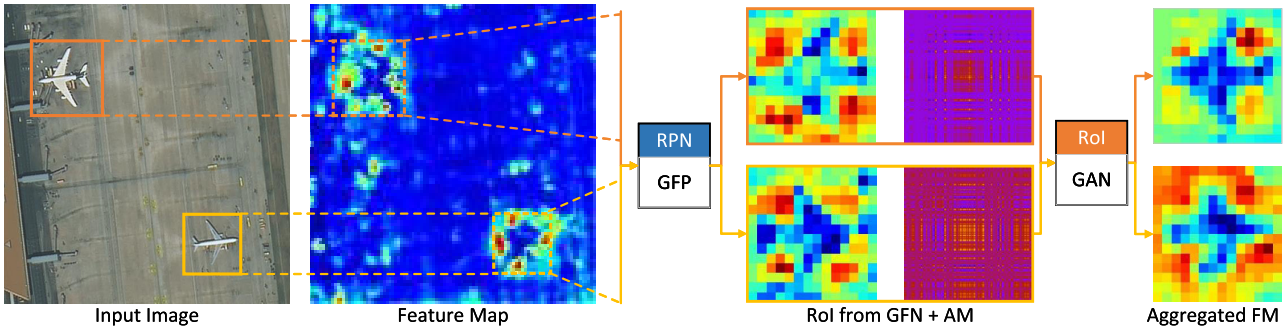


Fig. 15. Visualization of invariant structural features extracted based on graph modeling. From left to right: the input image, the feature map extracted by the backbone, the RoI and adjacency matrix generated by the GFP module, and the RoI obtained from the GAN.

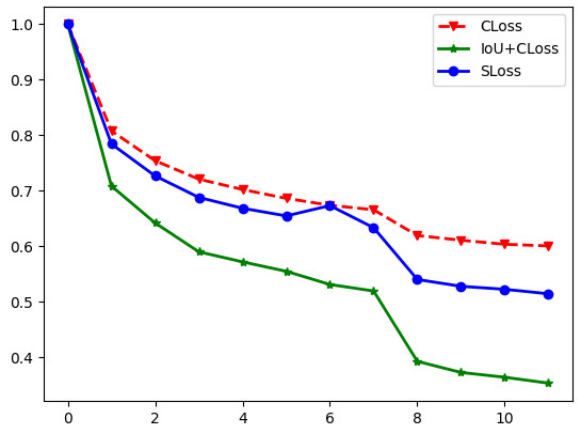


Fig. 16. We compare the three optional loss convergence curves applied in the first stage. Chamfer distance loss (CDL) represents only CD loss, IoU+CD loss is the final strategy utilized by the model, and SLoss represents a smooth L1 loss.

TABLE VI

COMPARISON OF GFN AND OTHER MAINSTREAM RPN-LIKE METHODS. □ INDICATES ANCHOR REPRESENTATION, + INDICATES DISTANCE REPRESENTATION, • INDICATES REPRESENTATIVE POINTS REPRESENTATION, AND ⋆ INDICATES EDGE POINTS REPRESENTATION. "RR" MEANS ROTATING REPRESENTATION PROVIDED

Method	Rep.	AR_{1000}	mAP	RR
RPN	□	45.7	65.8	
cascade RPN	□	46.8	66.9	
FCOS	+	46.6	68.5	
Fovea	+	46.5	68.6	
Rep	•	47.6	69.0	
GFN	⋆	**48.4**(+2.7)	**70.4**(+4.6)	✓

models utilize Resnet-50 with FPN as the backbone network. Our baseline model is a simple fusion of the heads of Faster-R-RCNN and Mask-RCNN. Different tasks utilize different annotations to guide the model for training and inference.

Combining the contents of Table IV and Fig. 14, we can draw the following conclusions.

1) Faster-RCNN (OBB) has poor performance in hard-distinguishable categories (B747 and ARJ21) and easily confused categories (B787 and A220).
2) The performance of the baseline of the merged two tasks in aircraft fine-grained recognition is slightly improved, which is mainly reflected in the accuracy improvement of the two hard-distinguishable categories. However, it is still not satisfactory in the easily confused categories;
3) GFA-Net surpasses all methods in these two tasks, not only further improving the accuracy of hard-distinguishable categories but also reducing the misjudgment of easily confused categories to a certain extent.

D. Ablation Study

Fig. 15 shows the visualization results of the feature maps of each stage from GFA-Net while inferencing images. We can initially derive the following two conclusions: 1) GFP can effectively extract the contour features of objects and 2) GAN aggregates features response between each node so that the overall result tends to be stable. The ablation experiments for each module are arranged as follows.

1) Graph Focusing Network: We propose the GFN to replace the traditional RPN structure. In remote sensing scenes, objects are a kind of symmetric topological structure with arbitrary-oriented and various scales. Traditional RPN encodes the object as a horizontal axis-aligned box, which will lead to the feature misalignment problem and further affect the quality of the proposals, while GFN pays more attention to extract the contour, specifically corners of objects, so as to more reasonably handle the RS objects. We design the following ablation experiment to verify the effectiveness of our proposed GFN.

As shown in Table V, we mainly select four aspects to conduct performance tuning for GFN. We have done experiments for feature maps between P_{2-6} and P_{3-7}, and the differences are downsampled ratios of these two. Since P_{2-6} is closer to the input layer compared to P_{3-7}, feature maps from P_{2-6} contain finer edge information, and feature maps from P_{2-6} contain richer semantic information. The edge information is more conducive to the model to distinguish the foreground and background, explaining the experimental results as well. Concerning the choice of the classification loss, we contrast cross-entropy loss and focal loss. In the case of similar results, we choose the cross-entropy loss (consistent with the baseline).

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TABLE VII

EXPERIMENTS FOR MODULE ABLATION AND PARAMETER TUNING ON GAN. WHEN OBSERVING THE MODULE ABLATION EXPERIMENTS, THE CORRESPONDING HYPERPARAMETERS NOT MENTIONED ARE CONFIGURED AS THE LAST LINE EXPERIMENT AND VICE VERSA. THE EXPERIMENTS ARE CONDUCTED ON THE DOTA DATASET

	Module			Metrcs		Hyperparameter			Metrics	
Method	Adjacency matrix	Fusion factor	Averaging Layer	mAP	FPS	Layer num.	Head num.	Channel	mAP	FPS
baseline	-	-	-	64.2	9.1	2	-	128	64.2	9.1
GAN				67.2		1	8	128	68.7	
	✓			68.2		3	8	128	71.5	
		✓		68.0		2	-	128	68.9	
			✓	67.8		2	4	128	70.2	
	✓	✓		70.6		2	16	128	71.4	
		✓	✓	69.9		2	8	64	70.4	
	✓		✓	68.7		2	8	256	71.5	
GAN	✓	✓	✓	71.3	11.2	2	8	128	71.3	11.2

TABLE VIII

COMPARISONS OF GRAPH FOCUSING ALGORITHM APPLYING DIFFERENT NODES SELECTION STRATEGY. MAP AND FPS ARE ADOPTED FOR PERFORMANCE TRADEOFF METRICS

Method	Strategy	classifier	mAP	FPS
FR-R	-	FC	65.8	8.8
GFA	A	baseline	65.2(-0.6)	14.6
		GAN	69.8(+4.0)	13.7
GFA	B	baseline	65.4(-0.4)	14.6
		GAN	69.9(+4.1)	13.8
GFA	C	baseline	66.7(+0.9)	13.9
		GAN	70.3(+4.5)	12.7
GFA	D	baseline	67.1(+1.3)	13.6
		GAN	70.1(+4.3)	12.4
GFA	$C+D$	baseline	68.1(+2.3)	13.3
		GAN	71.3(+5.5)	11.2

TABLE IX

COMPARISONS OF GFA-NET APPLYING DIFFERENT CLASSIFIERS. THE EXPERIMENTAL RESULTS ARE CONDUCTED ON THE DOTA DATASET

Method	Channel	Resolution	$AP^m_{0.5}$	$AP^m_{0.75}$	FPS
FC	1024	7×7	65.8	45.6	8.8
		14×14	66.3	46.1	
conv.	256	7×7	65.4	45.9	
		14×14	66.6	46.7	
GCN	128	7×7	64.2	41.8	9.1
		14×14	65.0	42.6	
GAT	128	7×7	67.9	49.8	10.1
		14×14	68.2	50.3	
GAN	128	7×7	71.3	52.3	11.2
		14×14	72.4	53.4	

Regarding the loss of localization, we also provide two options. The smooth L1 loss is used to learn the 4-D representation of the axis-aligned box, while IoU loss plus CD loss is used to locate the contour of an object. The experimental results in Fig. 16 show that: 1) CD loss with IoU loss has better convergence than CD loss itself, which shows that IoU loss has a certain auxiliary ability to CD loss; 2) IoU+CD loss is relatively more stable than smooth L1 loss, and the convergence result is better; and 3) the effective descending CD demonstrates the model's learning ability of the object contour at the first stage and further reflects the sufficient invariant structural feature extraction ability.

After completing the hyperparameters tuning of GFN, we also design a series of RPN-like structures based on the heads of several mainstream single-stage models. We ensure that irrelevant conditions are the same and only control the different representation methods of the predicted objects. As shown in Table VI, with the refinement of the object representation, the distinguishing capability of the model in the foreground and background is gradually enhanced, further optimizing the detection results. Especially in the point-based method, the performance of GFN applying corner points is better than that of reppoints [54] applying representative points, surpassing all other RPN-like performances. We analyze the main reasons for the performance improvement as follows: 1) GFN learns the corner points of objects, which is more interpretable than other methods (anchor-based, distance-based, and representation-based) and 2) compared with learning the axis-aligned RoI, GFN obtains a finer rotating RoI. On the one hand, the quality of the proposals is improved. On the other hand, the difficulty of the second-stage localization learning is reduced. The above comparison strongly illustrates the importance of fine-grained feature extraction for foreground proposals.

2) Graph Focusing Algorithm: The graph focusing algorithm is designed to integrate the edge information output

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by GFN, thereby assigning a focusing weight to each element on the RoI features. After obtaining the focus weight, the next step is to design the connection rules between nodes in the graph structure. In other words, an assignment strategy of the adjacency matrix A is necessary. We have designed four assignment strategies for matrix A, as shown in Fig. 8. The corresponding ablation experiment is given as follows.

We compare the mAP and FPS of our model applying baseline (graph convolution network) and GAN under four proposed strategies. Due to the relatively simple operation of the baseline in the node weight update process, it shows a higher speed in comparison with the performance of GFA-Net but an inferior performance (0.9 FPS higher but 3.7 mAP lower on average). As indicated in Table VIII, strategies A and B are similar in performance, but they both perform lower than strategy C. It can be seen that, in the manual-designed strategy, the relationship between the model performance and the neighbor nodes connection is weak but positively related to the number of connections. Strategy D and strategy C have achieved similar considerable performance. By adding strategy C to that strategy D, the performance can be further improved. We believe that the reason is that the connection of the nodes learned by the model is relatively irregular and not fixed. It helps to capture the relationship between medium and long-distance nodes and complements the artificially set regular connection. In general, for GAN, strategy D increases mAP by 1.5 on the basis of strategy A, and FPS drops by 2.5, which is acceptable.

3) Graph Aggregation Network: GAN is applied to aggregate the graph features output from the GFP, aiming to derive more clear contextual information. In this section, we first perform module ablation and hyperparameter tuning for the proposed GAN and then compare it with several classic methods.

We initially divide the ablation experiments in Table VII into two parts: modules and hyperparameters. The first and last lines correspond to the baseline model [graph convolution networks (GCNs)] and the proposed GAN (tuning completed), respectively. When observing the module ablation results, the default hyperparameters are set to the optimal case (last row). In turn, when looking at hyperparameter ablation results, the listed modules are all added by default.

As shown in Table VII, the adjacency matrix provides the focusing area of RoI features for GAN. If not applied, it means updating the features of all nodes. The fusion factor is used to measure the degree of participation of the neighbor nodes of each node. If not applied, it means that the importance is equivalent. We provide two versions of the GAL (concatenation and averaging) and utilize the averaging operation in the last layer. The experimental results verify the effectiveness of the above modules. In the module ablation part, we conduct the experiments of adjacency matrix, fusion factor, and average layer addition, respectively. Theoretically, the adjacency matrix is responsible for constructing graph nodes. The fusion factor is applied to weigh the importance measured between nodes. The average layer comprehensively considers the weight of each node. Functionally, the three do not interfere with each other and promote together. The experimental results show that the three parts improve the performance of the model independently of each other, which verifies our hypothesis. In the hyperparameter tuning part, we keep the number of layers and channels consistent with the baseline. At the same time, we have introduced a multihead mechanism, and the number is 8. Experimental results show that our GAN surpasses the baseline method (GCN) in terms of accuracy and speed.

After completing the hyperparameter tuning, we compare the performance of GAN with other methods. We conduct comparative experiments on RoI of two scales, respectively. According to Table IX, the following conclusions can be drawn: 1) the graph modeling layers and the convolution layer can obtain better performance on feature maps with higher resolution compared with the fully connected layer; 2) under the IoU matching thresholds of 0.5 and 0.7 (more stringent), GAN is still superior to other methods, showing strong localization accuracy; and 3) the number of channels applied by GAN is less than that of traditional methods, deriving a higher speed increase. Our analysis of the above experimental conclusions is given as follows: 1) compared with the fully connected layer to reconstruct position information in 1-D space, the convolutional layer and graph network layer methods can more efficiently utilize higher-resolution 2-D RoI feature maps; 2) our graph network modeling method extracts the contour information of objects relatively more completely with efficient node connection; and 3) using a relatively small number of channels, on the premise of stable performance, on the one hand, computing resources are saved, and on the other hand, the model inference speed is improved.

V. Conclusion

In general, we propose a remote sensing rotating fine-grained object detection model called GFA-Net. It defines a remote sensing object as a rigid structure composed of a set of unified components and adopts graph structure modeling and graph network for feature extraction. Compared with the traditional fully connected layers or fully convolutional networks, the graph network under graph structure modeling can achieve higher performance improvement with relatively fewer model parameters. Specifically, we design the GFP to complete the construction process of the object contour detection to the graph structure. We also propose a GAN to perform feature aggregation and extraction of the obtained graph structure. On the large-scale remote sensing fine-grained object detection dataset FAIR1M, the model's ability to localize and recognize fine-grained objects is verified. The ablation experiment on the public dataset DOTA verifies the effectiveness of each module, and the comparative experiments on DOTA and HRSC2016 illustrate the advantages of our model. It is confirmed on the proposed aircraft fine-grained dual-task dataset ACSD that, by introducing additional component information, the fine-grained recognition capability of GFA-Net can be further improved.

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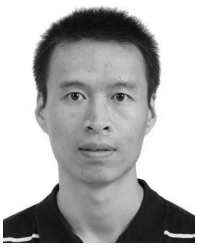
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